Tracking over time

Mean corrected error

Time

Tuned forgetting
Causal velocity
Exact motion control

Before and after correction

Corrected error

One-step forecast error

Repeated-data error

Identity
Tuned forgetting
Rolling window
Causal velocity
Exact motion control

Terminal chordal error

State uncertainty

Actual modal error

Nominal 95% posterior-ball radius

Fast drift, sensor-fixed inputs, n=8, rank 3. 64 independent trajectories; shaded bands are pointwise 95% intervals for mean error. Credible pilot: 8 trajectories; above diagonal means missed state. Open points flag failed sampling checks.